

Q1 2024 Earnings Call

Company Participants

- Mark Murphy, Executive Vice President and Chief Financial Officer
- Sanjay Mehrotra, President and Chief Executive Officer
- Satya Kumar, Corporate Vice President, Investor Relations and Treasury

Other Participants

- Aaron Rakers, Analyst, Wells Fargo
- Harlan Sur, Analyst, J.P. Morgan
- Krish Sankar, Analyst, TD Cowen
- Mehdi Hosseini, Analyst, Susquehanna Financial
- Timothy Arcuri, Analyst, UBS
- Toshiya Hari, Analyst, Goldman Sachs

Presentation

Operator

Thank you for standing by, and welcome to Micron Technology's First Quarter 2024 Financial Results Conference Call. At this time, all participants are in listen-only mode. After the speakers' presentation, there will be a question-and-answer session. (Operator Instructions). As a reminder, today's program is being recorded.

And now I'd like to introduce your host for today's program, Satya Kumar, Investor Relations. Please go ahead.

Satya Kumar {[BIO](#) [18241786](#) <GO>}

Thank you, and welcome to Micron Technology's Fiscal First Quarter 2024 Financial Conference Call. On the call with me today are Sanjay Mehrotra, our President and CEO; and Mark Murphy, our CFO. Today's call is being webcast from our Investor Relations site at investors.micron.com, including audio and slides. In addition, the press release detailing our quarterly results has been posted on the website, along with the prepared remarks for this call.

Today's discussion of financial results is presented on a non-GAAP financial basis unless otherwise specified. A reconciliation of GAAP to non-GAAP financial measures can be found on our website. We encourage you to visit our website at micron.com throughout the quarter for the most current information on the company, including information on financial conferences that we may be attending. You can also follow us on X at MicronTech.

As a reminder, the matters we are discussing today include forward-looking statements regarding market demand and supply, market trends and drivers, our expected results and guidance, and other matters. These forward-looking statements are subject to risks and uncertainties that may cause actual results to differ materially from statements made today. We refer you to the documents we file with the SEC, including our most recent Form 10-K and upcoming 10-Q, for a discussion of risks that may affect our future results. Although we believe that the expectations reflected in the forward-looking statements are reasonable, we cannot guarantee future results, levels of activity, performance, or achievements. We are under no duty to update any of the forward-looking statements to conform these statements to actual results.

I'll now turn the call over to Sanjay.

Sanjay Mehrotra {BIO 1977912 <GO>}

Thank you, Satya. Good afternoon, everyone. In fiscal Q1, Micron delivered revenue, gross margin, and EPS above the high-end of the guidance ranges we provided at the last earnings call, reflecting Micron's strong execution combined with improved pricing. We are in the very early stages of a multi-year growth phase catalyzed and driven by generative AI, and this disruptive technology will eventually transform every aspect of business and society. Memory is at the heart of GPU-enabled AI servers, and we are already seeing strong demand driven by early deployment of AI solutions, which will only accelerate over time. Micron is well positioned to leverage this growth, having executed the most robust set of new technology and product introductions in our 45-year history.

The improved supply-demand environment in the current calendar quarter gives us additional confidence in the trajectory of our business. We have driven a strong inflection in industry pricing this calendar quarter, which will allow us to benefit from higher prices earlier in our fiscal year compared to prior plans. We intend to stay very disciplined with our supply and capacity investments as our pricing is still far from levels associated with necessary ROI. We expect our pricing to continue to strengthen through the course of calendar 2024. We expect improved margins and financial performance throughout 2024, and record industry TAM in calendar 2025.

We have made significant progress with our industry-leading technology roadmap. Micron is at the forefront of ramping the industry's most advanced technology nodes in both DRAM and NAND. The vast majority of our bits are on leading-edge nodes: 1-alpha and 1-beta in DRAM and 176-layer and 232-layer in NAND. As previously stated, both 1-beta DRAM and 232-layer NAND nodes have reached mature yields faster than the prior nodes. We expect fiscal 2024 front-end cost reductions to track in line with our long-term expectations of mid- to high-single-digits in DRAM and low-teens in NAND. We are on track for volume production in 1-gamma DRAM using EUV in calendar 2025.

Now turning to our end markets. Inventories for memory and storage are at or near normal levels for most customers across PC, mobile, auto, and industrial end markets. Consequently, the demand that we see from customers in these markets is closer to their end-market demand. Data center customer inventory of memory and storage is improving, and we continue to expect customer inventory to approach normal levels in this market sometime in the first half of calendar 2024. Across our data center and PC markets, we are ahead of the industry in our transition to D5 and we expect to cross over our D5 volume from D4 in early calendar 2024.

Generative AI use cases are expanding from the data center to the edge, with several recent announcements of AI-enabled PCs, smartphones with on-device AI capabilities, as well as embedded AI in the auto and industrial end markets. The proliferation of on-device AI at the edge offers a host of benefits such as enhanced privacy, lower latency, improved performance, greater personalization and competitive costs for a wide range of use cases, from content creation to productivity. We see a rapid evolution in our customer product roadmaps enabling and leveraging this AI market expansion, which in turn is driving higher capacity, lower power, and increased performance requirements for memory and storage. We expect to increasingly benefit from content growth as these trends in AI gain momentum.

In data center, total server unit shipments are expected to increase by a mid-single-digit percentage in calendar 2024, following a year of low-double-digit percentage decline in calendar 2023. Demand for AI servers has been strong as data center infrastructure operators shift budgets from traditional servers to more content-rich AI servers. Also, in response to AI-driven

data center demand, several customers have announced aggressive roadmaps for new GPU and AI accelerator ASIC product introductions with increasing requirements for HBM capacity, performance, and power.

Micron is addressing these exciting opportunities brought on by the proliferation of AI with an industry-leading portfolio of data center solutions, including HBM3E, D5, several types of high-capacity server memory modules, LPDRAM, and data center SSDs. We have received very positive customer feedback on our HBM3E, which has approximately 10% better performance and about 30% lower power consumption compared to competitive offerings of HBM3E. In fiscal Q1, we shipped samples of HBM3E to a number of key partners and are making good progress in our qualifications. Micron is in the final stages of qualifying our industry-leading HBM3E to be used in NVIDIA's next-generation Grace Hopper GH200 and H200 platforms. In addition, our LP5x is being used for the Grace CPU, driving a new use case for LP memory in the data center for accelerated computing.

We are on track to begin our HBM3E volume production ramp in early calendar 2024 and to generate several hundred millions of dollars of HBM revenue in fiscal 2024. We expect continued HBM revenue growth in 2025, and we continue to expect that our HBM market share will match our overall DRAM bit share some time in calendar 2025.

Last month we introduced the industry's fastest and lowest latency 128GB high-capacity modules built on our industry-leading 1-beta node and using a monolithic die, which does not require 3D stacking and thus enables a simpler process flow for assembly. Featuring best-in-class performance, our solution will support customers' memory-intensive data center workloads today and into the future.

Additionally, leading CPU vendors have confirmed validation support for our monolithic-die-based 128GB modules on existing platforms released in 2022 and 2023 as well as upcoming new platforms. This ensures that our offering has a significant TAM that we can address immediately. We expect volume production to start next quarter, with significant growth in fiscal 2025 and beyond.

A testament to our solid execution and superior offerings, Micron ended the third calendar quarter with record-high revenue share in data center SSDs based on independent industry assessments. This marks the second consecutive quarter of record revenue share in data center SSDs, and we look to build on this revenue momentum through fiscal 2024.

In PCs, we forecast unit volumes to grow by a low- to mid-single-digit percentage in calendar 2024, after two years of double-digit percentage PC unit volume declines. We expect PC OEMs to start ramping AI-on-device PCs in the second half of calendar 2024, with an additional capacity of 4GB to 8GB of DRAM per unit, and we see average SSD capacities increasing as well. We also completed qualifications for our industry-leading 1-beta-based 16Gb D5 at several PC customers in fiscal Q1.

In fiscal Q1, we achieved record bit shipments in both client and consumer SSDs, as customers adopted our industry-leading solutions. Building upon our QLC leadership, our client SSD QLC bit shipments also reached a new record in fiscal Q1. QLC now comprises the majority of our bit shipment mix for both client and consumer SSD.

This month, we also announced that we are shipping the Micron 3500 NVMe SSD, the world's first performance client SSD with 200 plus layer NAND. Built on our industry-leading 232-layer NAND, the 3500 will help our customers handle demanding workloads for business applications, scientific computing, gaming, and content creation.

In mobile, smartphone demand is showing signs of recovery, and we forecast smartphone unit shipments to grow modestly in calendar 2024. Leading chipset vendors have announced powerful new products supporting on-device large language models with 10 billion or more parameters. We expect smartphone OEMs to start ramping AI-enabled smartphones in 2024, with an additional capacity of 4GB to 8GB of DRAM per unit.

Longer-term, many popular generative AI applications will be on smartphones, and our leading product portfolio is poised to capture this memory and storage opportunity. Our new industry-leading 9.6Gbps LP5x will address the bandwidth requirements of the most demanding AI-based mobile applications. We also began sampling our next-generation 232-layer NAND UFS 3.1 and our 1-beta DRAM 24Gb LP5x to support the memory needs of emerging AI foundational models.

Last, I'll cover auto and industrial, which are end markets we value as part of the portfolio due to their relatively more predictable revenue and profitability and long-term growth opportunity. The proliferation of AI at the edge continues to increase in the industrial and auto markets. For memory, this translates to content growth in a host of AI-enabled edge devices. For example, AI-enabled industrial PCs have 3x to 5x more memory than standard PCs and there is an 8x increase in memory content for AI-enabled edge video security cameras compared to standard non-AI video cameras.

Our automotive business achieved a new quarterly revenue record in fiscal Q1, driven by better demand and volume ramps of new vehicle platforms. As a leader in automotive market share and quality, Micron will benefit from memory and storage content growth as automotive OEMs expand features in ADAS and in-cabin applications. Our automotive design win trajectory remains strong.

Our industrial business saw double-digit sequential growth in fiscal Q1 as the industrial market continued to recover. Inventory levels for memory and storage continued to improve at distribution partners and are at normal levels at the majority of our customers. Industry fundamentals remain strong for memory and storage as the widespread adoption of IoT, AI, and machine learning solutions creates new growth opportunities for us.

Now turning to our market outlook, starting with demand. We expect calendar 2023 DRAM bit demand to grow in the high-single-digit percentage range, up from prior expectations for mid-single-digit growth. In NAND, we continue to expect calendar 2023 bit demand growth in the high-teens percentage range. Looking forward, over the next few years, we expect bit demand growth CAGRs of mid-teens in DRAM and low-20s percentage range in NAND. We forecast calendar 2024 bit demand growth for the industry to be near the long-term CAGR for DRAM and somewhat below the long-term CAGR for NAND.

Turning to supply, significant supply reductions across the industry have enabled the recovery that is now underway. An extended period of supply growth less than demand growth would strengthen the pace of recovery. Micron will continue to exercise supply and CapEx discipline, aligned with our strategy to maintain our long-term bit market share for DRAM and NAND. Micron's fiscal 2024 CapEx is projected to be between \$7.5 billion and \$8 billion, slightly higher than last year's levels and prior plans, primarily to support the HBM3E production ramp. We continue to expect WFE CapEx in fiscal 2024 to be down year-over-year.

As we have discussed previously, the ramp of HBM production will constrain supply growth in non-HBM products and will help improve the overall DRAM industry supply-demand balance. Across the industry, the HBM3E die is roughly twice the size of equivalent capacity D5. Additionally, the HBM product includes a logic interface die and has a substantially more complex packaging stack

that impacts yields. These factors result in HBM consuming more than 2 times the wafer supply as D5 to produce a given number of bits.

In last quarter's earnings call, we communicated that we strategically diverted underutilized equipment toward ramping new technology nodes, which will help us increase leading-edge production in a capital-efficient manner. Since the number of wafer processing steps is higher for leading-edge nodes, this approach of diverting underutilized tools to the leading edge meaningfully reduces our overall wafer capacity. Thus, underutilization in our fabs early this fiscal year transitions to structurally lower wafer capacity at higher utilization rates as we move through the fiscal year. Reports indicate that this redeployment of underutilized tools at the leading edge is an industry-wide practice that is likely to constrain industry supply in 2024.

Taking all these factors into account, Micron's bit supply growth in fiscal 2024 is planned to be well below demand growth for both DRAM and NAND, and we expect to decrease our days of inventory in fiscal year 2024. We expect calendar 2024 industry supply to be below demand for both DRAM and NAND, which will result in a contraction of industry inventory levels. As we have highlighted before, we continue to work with the US government and CHIPS grants are assumed in our CapEx plans for fiscal 2024. The viability and global competitiveness of our Idaho and New York projects depends on Micron receiving CHIPS grants to address the cost difference compared to overseas expansion.

To better support our customers around the globe, we have opened state-of-the-art assembly and test facilities in Malaysia and Taiwan. We are proceeding with our previously announced expansion of our Xi'an facility, having received approval from Chinese authorities for our planned investment. In fiscal Q1, we achieved the first mobile customer qualification of LPDRAM assembled at our Xi'an site, furthering our strong commitment to serve our mobile customers in China.

Our broad, diverse network of global operations remains a key element of our strategy to address customer demand in a reliable and resilient fashion. Our leading technology, strengthening product portfolio, strong manufacturing capabilities, and our dedicated team members position us well to capture the opportunities ahead.

I will now turn it over to Mark for our financial results and outlook.

Mark Murphy {[BIO 16136048](#) <GO>}

Thanks, Sanjay, and good afternoon, everyone. Micron delivered strong results in fiscal Q1 with revenue, gross margin, and EPS higher than the upper end of the guidance range we provided in our last earnings call. During the quarter, an improving supply-demand environment and our team's strong execution resulted in higher prices across DRAM and NAND. The current pricing trajectory has improved our financial outlook for the second quarter and full fiscal year.

Total fiscal Q1 revenue was approximately \$4.7 billion, up 18% sequentially and up 16% year-over-year. Fiscal Q1 DRAM revenue was \$3.4 billion, representing 73% of total revenue. DRAM revenue increased 24% sequentially, with bit shipments increasing in the low-20s percentage range and prices increasing in the low-single-digit percentage range.

Robust bookings entering the quarter, including customer strategic buys that occurred in prior quarters for fiscal Q1 shipment, limited our reported fiscal Q1 DRAM price increases despite Micron's strong execution on CQ4 pricing. Our strong calendar Q4 price execution contributes to our solid sequential growth in fiscal Q2, even with the effect of seasonality.

Fiscal Q1 NAND revenue was \$1.2 billion, representing 26% of Micron's total revenue. NAND revenue increased 2% sequentially with pricing more than offsetting an expected and communicated decline in volumes. Bit shipments declined in the mid-teens percentage range after record shipments in the prior quarter, and prices increased by approximately 20%. Portfolio mix improvements in NAND contributed to the increase.

Now turning to revenue by business unit. Compute and Networking Business Unit revenue was \$1.7 billion, up 45% sequentially. Data center and client shipments strengthened in the quarter. AI-related shipments increased in the data center market and normalized inventory at client customers enabled bit shipment growth.

Revenue for the Mobile Business Unit was \$1.3 billion, up 7% sequentially. Mobile revenue continued to show strength as customer inventories normalized and smartphone units and average memory and storage capacity growth at customers drove demand. Our mobile fiscal Q1 revenue almost doubled from year-ago levels.

Embedded Business Unit revenue was \$1 billion, up 21% sequentially. Growth was strong across most end markets. Revenue for the Storage Business Unit was \$653 million, down 12% sequentially due to sharply lower consumer component sales, partially offset by strong growth in SSD revenue.

The consolidated gross margin for fiscal Q1 was near 1%, improving 10 percentage points sequentially and driven by higher prices and a greater mix of DRAM products. Operating expenses in fiscal Q1 were \$992 million, up \$150 million sequentially and in line with our late November update. OpEx increased on higher R&D expenditures and the reinstatement of certain compensation programs suspended in the prior fiscal year including short-term incentive compensation.

We had an operating loss of \$955 million in fiscal Q1, resulting in an operating margin of negative 20%, improved from negative 30% in the prior quarter. Fiscal Q1 taxes were \$59 million, lower than the anticipated \$80 million based on an updated view of projected taxes across the year, driven by our improved fiscal 2024 outlook. The non-GAAP loss per share in fiscal Q1 was \$0.95 compared to a non-GAAP loss per share of \$1.07 in the prior quarter and a non-GAAP loss per share of \$0.04 in the year-ago quarter.

Turning to cash flows and capital spending, our operating cash flows were approximately \$1.4 billion in fiscal Q1 representing 30% of revenue. During the quarter, we received \$600 million in customer prepayments to secure supply for leading-edge memory products. Capital expenditures were \$1.7 billion during the quarter resulting in free cash flow of negative \$333 million in the quarter.

Our fiscal Q1 ending inventory was \$8.3 billion or 159 days, down from 170 days in the prior quarter. As mentioned in prior quarters, we hold strategic inventory stock associated with build ahead of products for cost optimization and risk mitigation. Excluding strategic stock, our fiscal Q1 ending inventory days would be approximately 142 days, only 22 days higher than our target inventory level.

On the balance sheet, we held \$9.8 billion of cash and investments at quarter end and maintained \$12.3 billion of liquidity when including our untapped credit facility. We ended the quarter with \$13.5 billion in total debt, low net leverage, and a weighted average maturity on our debt of 2030.

Now turning to our outlook for the fiscal second quarter. While we remain mindful of macroeconomic risks, the memory and storage market environment is improving. We expect supply-demand balance to tighten in both DRAM and NAND throughout 2024. Our leading-edge DRAM and NAND nodes are oversubscribed for the full year. Consequently, we expect prices to increase through calendar 2024, driving improvements in our financial performance.

Our leading-edge inventory is very tight, and we are also working to minimize pull-in of customer demand in response to higher pricing. As a result, our sequential growth in the near term will be driven primarily by pricing rather than a sequential increase in bit shipments. Both DRAM and NAND bit shipments are expected to decline somewhat in the fiscal second quarter.

We expect our fiscal Q2 gross margin to benefit from sequential price increases and reduced impact from underutilization. We project the balance of previously written-down inventories to clear in fiscal Q2. We forecast operating expenses to decline in the fiscal second quarter on lower R&D program expenses and an asset sale previously expected to occur in the first quarter. For the fiscal year, due to some higher R&D expenses including what we saw in Q1 and higher short-term incentive compensation from an improved outlook, we now project OpEx to be over \$3.9 billion. We forecast a much reduced operating loss in fiscal Q2 and project a return to operating income in Q3.

Our tax forecast for the year has increased from under \$200 million to over \$300 million based on an updated taxable income outlook. We project the allocation of tax expense across the year to be heaviest in the fourth quarter driven by profitability and other factors. As we've mentioned previously, at current levels of profitability, tax estimates and the distribution of taxes across the year are highly sensitive to changes in the outlook. We plan fiscal Q2 capital expenditures to be in line with first quarter levels. We see operating cash flows improving substantially in the second half of the fiscal year and are now forecasting positive free cash flow in the fiscal fourth quarter.

With all these factors in mind, our non-GAAP guidance for fiscal Q2 is as follows. We expect revenue to be \$5.3 billion, plus or minus \$200 million; gross margin to be in the range of 13%, plus or minus 150 basis points; and operating expenses to be approximately \$950 million, plus or minus \$15 million. We expect tax expenses of approximately \$45 million. Based on a share count of approximately 1.1 billion shares, we expect a loss of \$0.28 per share, plus or minus \$0.07.

In closing, the industry environment is improving, and our financial outlook has strengthened for the fiscal year and beyond. We will continue to take a disciplined approach to managing the business and remain focused on optimizing price, driving productivity, and controlling capital spend. With high levels of liquidity and low net leverage, we continue to operate from a position of balance sheet strength as we forecast a return to profitability and positive free cash flow.

I will now turn it back over to Sanjay.

Sanjay Mehrotra {[BIO](#) [1977912](#) <GO>}

Thank you, Mark. Over the last year, our world-class technology, business, and manufacturing teams ensured ongoing leadership in foundational memory technologies and the expansion of our industry-leading product portfolio. We are encouraged by the progress we have made on pricing, and we are on track to restore profitability more commensurate with the great value our solutions provide to our customers.

We expect 2024 to be a year of recovery and can see the path towards a healthy supply-demand environment along with strong growth in critical new technologies like HBM3E. From the data center to the edge, AI has emerged as a significant secular driver that will further bolster the

industry towards record revenue TAM in 2025 and drive growth for years to come. Micron's broad and growing suite of leading-edge products positions us well to capitalize on the immense opportunities ahead.

Thank you for joining us today. We will now open for questions.

Questions And Answers

Operator

Certainly. (Operator Instructions). One moment for our first question. And our first question comes from the line of Krish Sankar from TD Cowen. Your question, please.

Q - Krish Sankar {[BIO 16151788 <GO>](#)}

Yeah. Hi. Thanks for taking my question. Sanjay or Mark, the first question I had is kind of you spoke about sustainability of pricing in calendar '24. I'm kind of curious if you can peel the onion one layer below and say how we think about pricing through calendar '24 for DRAM and NAND, and if you can extrapolate it into 2025, that'll be very helpful. And then I had a follow-up.

A - Sanjay Mehrotra {[BIO 1977912 <GO>](#)}

Thanks, Krish for the question. And with respect to the pricing, we of course expect pricing to continue to strengthen during calendar 2024. And this is because of the healthy demand-supply balance as we discussed in the context of our script. As you have seen, there have been significant cuts in supply growth in the industry. Customer inventories have normalized, supplier inventories are improving, as we have discussed our own inventory here as well. And pricing will continue to improve as a result through the course of the year. And of course, you know, the demand trends overall, because of improvement of customer inventories are in PCs, in smartphones, automotive and industrial, the demand trend will continue. And in sometime in first half of '24, calendar '24, we expect data center inventories at customers to get normalized as well. And beyond that point, we would expect data center to become another boost in demand in 2024. So we expect pricing to continue to increase both in NAND and in DRAM as well. And we expect a healthy demand-supply environment in 2025 as well as a healthy pricing environment in '25 too.

And I just want to point out that as we noted, we have tightness on our leading-edge nodes. They are already in short supply and inventories will continue to improve for us. And all of this results in overall healthy dynamics for pricing improvements, profitability improvements, and revenue opportunity growth in the backdrop of demand drivers, AI being a dominant demand driver across the end markets.

Q - Krish Sankar {[BIO 16151788 <GO>](#)}

Got it. That's very helpful, Sanjay. And then just a quick follow-up on CapEx. I understand a lot of the CapEx is going towards HBM and WFE is going to be down year-over-year. Is there a way of quantifying how much of the CapEx for next year is for HBM? And at -- what is the catalyst for you to start investing in NAND? Is it price? Is it gross margin? Any such catalyst for NAND. That'd be very helpful. Thank you.

A - Sanjay Mehrotra {[BIO 1977912 <GO>](#)}

So we are not breaking down the CapEx, you know, between HBM and other parts of the business, but we have noted that our WFE is down, and of course, we are very much focused on

supporting the growth of our HBM business. We are very excited about our leading-edge product, HBM3E, which is the most advanced HBM3E offering in the industry. And we are -- as we noted in our remarks, post the qualification, you know, we are going to be starting production ramp early in calendar '24 and driving revenue growth will be more back half loaded for us. So -- I mean, we will of course make the necessary investments to support the demand for our HBM in '24. I would just point out that our HBM supply is basically in calendar year '24 is sold out at this point.

And, you know, in terms of, you know, NAND and CapEx, whether in NAND or in DRAM, what's most important is that the profitability returns to the levels that are really needed in order to justify ROI in increasing any CapEx investments. So this is what we are focused on and we'll remain extremely disciplined with respect to any CapEx, with respect to any supply growth considerations. And you know that the profitability in the industry is still far from the levels that are needed to get ROI on the investments and we plan to remain extremely disciplined in this regard.

Q - Krish Sankar {[BIO 16151788](#) <GO>}

Thanks, Sanjay.

Operator

Thank you. One moment for our next question. And our next question comes from the line of Aaron Rakers from Wells Fargo. Your question, please.

Q - Aaron Rakers {[BIO 6649630](#) <GO>}

Yeah. Thanks for taking the question. I have two as well real quick. You know, first, building on the HBM comment that you're basically sold out for the year. I know that you referenced in your prepared remarks that you're qualified in with the GH200 and the H200 at NVIDIA. I'm just curious, when you get quals in on HBM3E, you know, how does that market work in terms of -- you know, do these customers dual qualify? Do you have line of sight and kind of your share position within those product skews? I'm just curious of how the qualifications work and the visibility that you have to basically kind of, you know, double down on the fact that you expect to be kind of, you know, comparable market share in HBM as DRAM in total as you look out into '25.

A - Sanjay Mehrotra {[BIO 1977912](#) <GO>}

So our product is in qualification, as we noted in our comments here, and qualification is progressing well. And post qualification, we expect to be ramping production volumes as well as shipments to our customers. And that's what will yield to several hundred millions of dollars of revenue for us in fiscal year '24. Again, that revenue will be more back half of our fiscal year, more so in the back half. But that volume ramp and that revenue opportunity will continue to build up as we go into '25. It will continue to increase in our fiscal year as well as calendar year '25.

And as we have said that sometime in calendar year '25, we expect to be able to get to a level of our share in the HBM market that would align with our share in DRAM. So this is an exciting opportunity for us. And as you know, HBM is higher revenue per gigabyte. It is also higher profitability per gigabyte. So this is -- and it's one of the biggest growth markets in memory today. So we are excited about this opportunity.

And of course, you know, there's a very tight relationship, tight integration with customers for quals on HBM. And it takes longer than standard products. And, you know, these are all factors

that will play a role in terms of, you know, the number of suppliers that customers would tend to have for any given platform for HBM.

Q - Aaron Rakers {[BIO 6649630](#) <GO>}

Yeah. And then as a quick follow-up, I'm curious, in the comment you had said that basically your leading-edge inventory was very tight. And I think the follow-up comment was that you're really, you know, focusing on minimizing any pull-in in the midst of, you know, pricing increases materializing. Can you just help us appreciate what exactly you're able to do to minimize any pull-in or the visibility you have in whether any customers are taking strategic inventory? Just curious how you manage that.

A - Sanjay Mehrotra {[BIO 1977912](#) <GO>}

So with respect to strategic inventory, I think we had discussed in the past that we had made certain strategic -- certain customers had built certain strategic purchases in our FQ4 time frame and those are completing in CQ4 here. And of course, as we look ahead, we are very much focused on, you know, managing our supply, managing our demand. Pricing environment is increasing and we want to in this environment, of course, make sure that we meet the requirements of our customers in a fashion that overall maintains, you know, healthy dynamics of our business. Beyond that, I don't think we are in a position to discuss further specifics here.

Q - Aaron Rakers {[BIO 6649630](#) <GO>}

Fair enough. Thank you.

Operator

Thank you. One moment for our next question. And our next question comes from the line of Toshiya Hari from Goldman Sachs. Your question, please.

Q - Toshiya Hari {[BIO 6770302](#) <GO>}

Hi. Good afternoon. I had two questions as well. One on gross margins, maybe for Mark, and then my follow-up is for Sanjay. Mark, so the gross margin profile is improving from, you know, breakeven or 1% in the prior quarter to 13%. I was hoping you could sort of break that down, the sequential change for us in terms of pricing, I guess lower underutilization charges, and whatever else is happening from an inventory perspective. And beyond Feb, as we sort of look out into May, you talked about pricing improving throughout '24, so that continues to be a tailwind. How should we think about some of the other dynamics that go into your gross margin math?

A - Mark Murphy {[BIO 16136048](#) <GO>}

Sure. Excuse me. So on the fourth quarter to the first quarter gross margin, you know, we were up 10 points over half, or about half of that was price. Most of the remainder was mix on the higher DRAM volumes. And then we did see some favorable cost. We had the lower-cost inventories clear at \$600 million and then we had some reduced idle charges that we've talked about.

Now, if we move out to second quarter, as we mentioned, we're not seeing volume growth in the second quarter, but we are seeing gross margins still up 12 points. So it's all driven by price, or principally price. There is some mix shift within, you know, by customers and some seasonal effects, but again, it's largely a price-driven increase. You know, while we have lower benefits from the low-cost inventory clearing, we'll have \$400 million clear or \$400 million of benefit in the second quarter when we had \$600 million in the first quarter. We are seeing some cost declines

occurring with the increase of leading node production, and then again with the lower wafer starts and the higher utilization. What we've talked about before, we start to see idle charges dropping, as we've discussed. So again, principally price in the second quarter, but then beginning to see some cost benefits, even though we're losing the benefit of that lower cost inventory.

We will see price appreciation through the year. We're going to -- we don't expect there to be volume growth in the third quarter either, but good price appreciation, which will drive gross margins up. And then in the fourth quarter, we would expect to see volume and price, and again some lower utilization charges. So again, we would expect to see margin expansion second quarter to third quarter, and then again third quarter to fourth quarter.

Q - Toshiya Hari {BIO 6770302 <GO>}

Okay. Great. That's super helpful. And then, as my follow-up for Sanjay, you know, you guys had presented a, you know, cross-cycle financial model at your Investor Day, I think it was last May. Clearly, the environment has changed quite a bit. But when you sort of, you know, talk about necessary ROI for your business, as you guys debate when to increase production, when to increase CapEx, should we look at the model from last May still as a reference point, you know, through cycle operating margins of 30%, free cash flow margins of 10% or higher? Is that still the model that's relevant in your view, or have things changed since? Thank you.

A - Sanjay Mehrotra {BIO 1977912 <GO>}

Yes, that model is very much relevant once we get past this downturn and the recovery from this downturn as well. And I think we just have to recognize that this downturn has been steep. This has been driven by once in 100 year pandemic as well as all the other related factors which we have talked about, customer inventories and demand pull-in and demand normalization and all other things that have impacted -- severely impacted the industry environment over the course of last year.

2024, we'll be recovering and we have called it the year of recovery. And we have talked about how we see pricing continuing to increase through the year and of course, profitability continuing to improve through the year as well. As we look past this recovery, we would definitely say that cross-cycle model that we have discussed in the past definitely would hold and that's what we target for.

Q - Toshiya Hari {BIO 6770302 <GO>}

Thank you.

Operator

Thank you. One moment for our next question. And our next question comes from the line of Timothy Arcuri from UBS. Your question, please.

Q - Timothy Arcuri {BIO 3824613 <GO>}

Thanks a lot. I had a question on these prepaids that you're getting. That's \$600 million. That's a pretty big number. So is that more of a one-time deal or should we expect these, you know, prepaids to continue? And as part of that, is that mostly related to say HBM or a particular vertical like data center or is that across, you know, most of your end markets? And then I had a follow-up as well.

A - Sanjay Mehrotra {[BIO 1977912 <GO>](#)}

So obviously, due to the confidential nature of this agreement, we cannot provide any specifics around these prepayments. But what I would like to point out is that it does reflect the importance of our technology, our products, and our delivery capabilities. It also reflects our close relationships with our partners and, you know, commitment from both sides from our -- a good example of commitment from customers as well as from Micron. Beyond that, I'm not able to really provide any specifics here. And again, honoring the confidentiality.

Q - Timothy Arcuri {[BIO 3824613 <GO>](#)}

Okay. Okay, got that. I guess then can you talk, Sanjay, just about limiting the bit shipments? I think you said you're limiting the bit shipments to prevent pull-ins ahead of price increases. Sounds like bits are flat sequentially for fiscal Q2 and fiscal Q3. Can you talk about the logistics of that? Are you just kind of holding back on volumes to regain some, you know, pricing leverage? I guess if your competitors don't match that approach, you might risk losing some shares. So can you just talk about the logistics of that? Thanks.

A - Sanjay Mehrotra {[BIO 1977912 <GO>](#)}

Well, as we noted that leading-edge supply is already tight. And, you know, so that, you know, certainly impacts in our FQ2, some of the shipments. FQ2, of course, is also impacted by seasonality. And -- so supply is -- managing supply, given the tight environment of supply on the leading nodes, is really the main consideration in terms of us guiding you to this profile. And of course, as we manage, as we allocate that supply across the customers, we want to make sure that we are managing our shipments to our customers carefully.

Q - Timothy Arcuri {[BIO 3824613 <GO>](#)}

Okay. Thank you so much.

Operator

Thank you. One moment for our next question. Our next question comes from the line of Mehdi Hosseini from Susquehanna Financial. Your question, please.

Q - Mehdi Hosseini {[BIO 4362002 <GO>](#)}

Yes. Thanks for taking my question. Two follow ups. Sanjay, historically, memory industry tends to gravitate towards higher-margin products, which has historically led to margin erosion. Why is HBM any different? And what are your thoughts? Anything you can share with us as to what can preserve the higher margin associated with these high-end products? And I have a follow-up.

A - Sanjay Mehrotra {[BIO 1977912 <GO>](#)}

I think, you know, AI is in very early innings. Gen AI is barely starting and these are great growth opportunities ahead. You know, this is, you know, biggest revolution since Internet and recently you have heard industry estimates of data center AI accelerator TAM CAGR of being about 70% over the course of next few years. Of course, as those opportunities grow with data center accelerators, you know, from various suppliers, of course, the whole infrastructure grows and it's about AI and Gen AI applications from training to inferencing and really proliferating all across, you know, the data center environment. And that's where -- and along with the growth in data center AI accelerators, the rest of the infrastructure, including HBM, will continue to grow.

So we project that HBM CAGR will be over 50% over the course of next few years. And when you think about it, that is more than 3 times of the DRAM industry CAGR that we are talking about. And still, we are in the very, very early innings. 2023 is the first year of meaningful shipments of HBM in the industry and that too corresponds to low-single-digit percentage in terms of bits shipped in HBM this year, but much higher pricing, much higher revenue opportunity. So as we look ahead, we see HBM continuing to grow strongly in the industry. Its demand will grow. It will be a key enabler of Gen AI applications in training as well as inferencing because more and more data is required. As you look at more and more, larger and larger large language models, and more training on more data just drives more demand for high bandwidth, high performance, low power memory. So this is the very beginning. It has long ways to go.

And the other important factor with HBM, as we have discussed is, that it really takes more than 2 times as many wafers to produce the same number of bits as D5. So it really has -- it is a headwind to the supply growth in the DRAM industry and it has the effect of helping strengthen the supply-demand balance of the industry as well. So I think these are, you know, some of the important aspects.

Of course, when -- so important thing is that this has to be looked at as a long-term opportunity, long-term growth opportunity. And of course, you know, we are excited about getting our HBM share to align with our DRAM share some time in 2025. And of course, as we look at any large opportunity, you know, over time, it will certainly have some ebb and flow in terms of demand and supply and we will prudently manage this. And maintaining flexibility in managing this is absolutely key. And you have seen us manage this well over time in our overall industry for DRAM as well as for NAND on part of Micron, and we'll continue to manage it in that fashion. It is an exciting opportunity. We are well positioned with our product and we look forward to continuing to grow revenue and profit contribution with this product line over the course of next few years.

Q - Mehdi Hosseini {[BIO 4362002](#) <GO>}

Great. Thanks for detail. Just a quick follow-up for Mark. Is there a normalized capital intensity that we should think of, especially as we come off this kind of nuclear winter in memory? Is there any -- if you don't have a normalized capital intensity, what else out there that could help us better forecast free cash flows?

A - Mark Murphy {[BIO 16136048](#) <GO>}

Yeah, no update, Mehdi, to the cross-cycle model that we've provided. So I think the best way to think about CapEx is just over time as a percent of revenue, which we've given mid 30s percent over time. Of course, you know, for example, HBM, you know, requires more CapEx, but it also yields a price premium and accretive margin. So we believe that at this time that capital intensity model that we've provided before is still intact.

Q - Mehdi Hosseini {[BIO 4362002](#) <GO>}

Thank you.

Operator

Thank you. One moment for our next question. And our next question comes from the line of Harlan Sur from J.P. Morgan.

Q - Harlan Sur {[BIO 6539622](#) <GO>}

Good afternoon. Thanks for taking my question. As you mentioned, your leading-edge DRAM and NAND supply output is oversubscribed for the full year. Does you factor in your CapEx plans, conversion of capacity from mature nodes to leading-edge nodes? When do you guys see your ability to fully supply to your customers' demand profile? Is that looking more likely now calendar year '25 and what end markets or applications are you seeing the largest demand-supply gap as you move through the fiscal year?

A - Sanjay Mehrotra {BIO 1977912 <GO>}

As we move through the fiscal year, of course, you know, leading-edge nodes being in tight supply. You know, leading-edge nodes are, you know, driver of demand in PCs and smartphones and data center applications and we have a strong portfolio that is well positioned with these nodes. And -- so -- I mean, basically, this is important for us to maintain the supply discipline and we -- industry is still not at the profitability levels for investments to be made and we'll manage our supply very, very prudently as we work ahead. And of course, we'll continue to focus on driving the pricing as well as driving the profitability of our business to bring in return on investments on our CapEx that is needed and do our best to manage and allocate the demand across our end markets and customers.

Q - Harlan Sur {BIO 6539622 <GO>}

Great. Thanks for that. And again, you mentioned this briefly in your prepared remarks, but if you look at some of the third-party research estimates, September quarter was the second consecutive quarter where the team had, I think, somewhere about 10% to 12% market share in data center NVMe SSDs. If I look back historically, like your share has been more in the sort of 3% to 5% range. So obviously strong recent market share performance. You've got a strong lineup of data center SSDs. What's been the big differentiator for the team here and how do you guys continue to grow your share going forward?

A - Sanjay Mehrotra {BIO 1977912 <GO>}

So thank you for acknowledging that. I mean, our team has done a great job with data center SSD product portfolio over the course of the years and now we have a strong set of product offerings for, you know, data center SSDs, NVMe SSDs and we now achieved record data center market share in SSDs for two consecutive quarters. And for the calendar Q3, our data center SSD share now is in line with our NAND share in the industry. And this is really the benefit, and you're seeing the full benefit, the full power of vertical integration playing out here where our team has worked together from device to design to firmware to system implementation to understanding of customer application, working closely with customers in qualifications and really across a wide range of customers over course of time, have really developed very robust industry-leading, strong product portfolio with greater opportunities ahead as well.

So, this is the transformation that we began to drive in the company, going from selling components in the past to value-add solutions and really very pleased how data center SSD recognition is being provided to our team in terms of revenue opportunities by our customers and, you know, certainly reflected in the share gains that we have made in this market.

And of course, another big factor that has been a key contributor to the success here is you may recall we transitioned some time ago from floating gate to replacement gate technology in NAND and that has definitely been a key factor. Strong, successful, timely execution on that has played an important role in our data center SSD strength. So we are very excited about Micron's market position in this market and our future opportunities here.

Q - Harlan Sur {BIO 6539622 <GO>}

Thank you, Sanjay.

Operator

Thank you. This does conclude the question-and-answer session as well as today's program. Thank you, everyone, for participating in today's program. You may now disconnect. Good day.

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